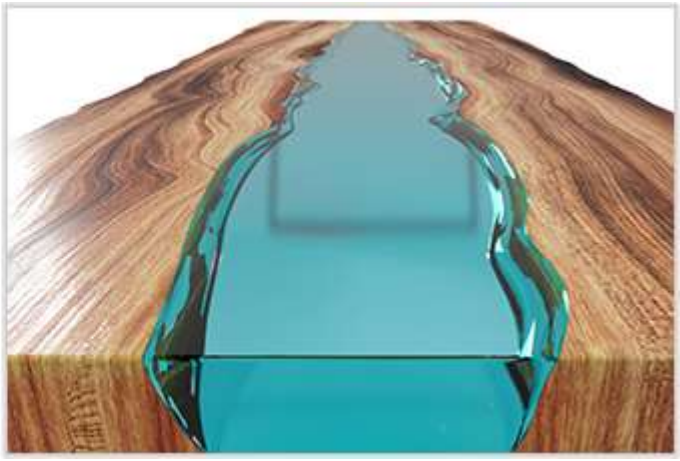
Description

Image





Caption

1. Epoxy resin poured to create a hardwearing floor surface. 2. Epoxy resin can be used creatively and decoratively in interior furnishings. Images used under license from Shutterstock.com

The material

Epoxies are thermosetting polymers with excellent mechanical, electrical and adhesive properties and good resistance to heat and chemical attack. They are used for adhesives (Araldite), surface coatings and, when filled with other materials such as glass or carbon fibers, as matrix resins in composite materials. Typically, as adhesives, epoxies are used for high-strength bonding of dissimilar materials; as coatings, they are used to encapsulate electrical coils and electronic components; when filled, they are used for tooling fixtures for low-volume molding of thermoplastics.

Compositional summary

(O-C6H4-CH3-C-CH3-C6H4)n

General properties

Density	1.11e3	-	1.4e3	kg/m^3
Price	* 1.9	-	3.8	GBP/kg
Date first used	1947			

Mechanical properties

Young's modulus	2.35	-	2.47	GPa
Shear modulus	* 0.84	-	0.883	GPa
Bulk modulus	* 3.87	-	4.06	GPa
Poisson's ratio	* 0.391	-	0.407	
Yield strength (elastic limit)	* 36	-	71.7	MPa
Tensile strength	45	-	89.6	MPa
Compressive strength	* 103	-	172	MPa

Elongation		3	-	6	% strain
Hardness - Vickers	*	11	-	22	HV
Fatigue strength at 10^7 cycles		30	-	35	MPa
Fracture toughness		0.57	-	0.63	MPa.m^0.5
Mechanical loss coefficient (tan delta)	*	0.0162	-	0.017	

Thermal properties

Glass temperature	*	66.9	-	167	°C
Maximum service temperature		122	-	138	°C
Minimum service temperature	*	-43.2	-	6.85	°C
Thermal conductor or insulator?		Good insulator			
Thermal conductivity		0.181	-	0.196	W/m.°C
Specific heat capacity		1.18e3	-	1.24e3	J/kg.°C
Thermal expansion coefficient		81	-	117	µstrain/°C

Electrical properties

Electrical conductor or insulator?		Good insulator			
Electrical resistivity		1e21	-	6e21	µohm.cm
Dielectric constant (relative permittivity)		3.9	-	4.1	
Dissipation factor (dielectric loss tangent)		0.007	-	0.008	
Dielectric strength (dielectric breakdown)		11.8	-	19.7	MV/m

Optical properties

Transparency		Transparent			
Refractive index		1.54	-	1.6	

Critical Materials Risk

High critical material risk?		No			

Processability

Castability		4	-	5	
Moldability		4	-	5	
Machinability		3	-	4	
Weldability		1			

Durability: water and aqueous solutions

Water (fresh)		Excellent			
Water (salt)		Excellent			
Soils, acidic (peat)		Limited use			
Soils, alkaline (clay)		Acceptable			

Values marked * are estimates.
ANSYS, Inc. provides no warranty for this data.

Wine	Limited use

Durability: acids

Acetic acid (10%)	Limited use
Acetic acid (glacial)	Unacceptable
Citric acid (10%)	Excellent
Hydrochloric acid (10%)	Excellent
Hydrochloric acid (36%)	Limited use
Hydrofluoric acid (40%)	Unacceptable
Nitric acid (10%)	Limited use
Nitric acid (70%)	Unacceptable
Phosphoric acid (10%)	Excellent
Phosphoric acid (85%)	Acceptable
Sulfuric acid (10%)	Excellent
Sulfuric acid (70%)	Limited use

Durability: alkalis

Sodium hydroxide (10%)	Limited use
Sodium hydroxide (60%)	Excellent

Durability: fuels, oils and solvents

Amyl acetate	Acceptable
Benzene	Acceptable
Carbon tetrachloride	Excellent
Chloroform	Unacceptable
Crude oil	Excellent
Diesel oil	Excellent
Lubricating oil	Excellent
Paraffin oil (kerosene)	Excellent
Petrol (gasoline)	Excellent
Silicone fluids	Excellent
Toluene	Acceptable
Turpentine	Acceptable
Vegetable oils (general)	Excellent
White spirit	Excellent

Durability: alcohols, aldehydes, ketones

Acetaldehyde	Unacceptable
Acetone	Unacceptable

Ethyl alcohol (ethanol)	Limited use
Ethylene glycol	Acceptable
Formaldehyde (40%)	Excellent
Glycerol	Excellent
Methyl alcohol (methanol)	Unacceptable

Durability: halogens and gases

Chlorine gas (dry)	Limited use
Fluorine (gas)	Unacceptable
O2 (oxygen gas)	Unacceptable
Sulfur dioxide (gas)	Excellent

Durability: built environments

Industrial atmosphere	Excellent
Rural atmosphere	Excellent
Marine atmosphere	Excellent
UV radiation (sunlight)	Fair

Durability: flammability

Flammability	Slow-burning

Durability: thermal environments

Tolerance to cryogenic temperatures	Unacceptable
Tolerance up to 150 C (302 F)	Excellent
Tolerance up to 250 C (482 F)	Limited use
Tolerance up to 450 C (842 F)	Unacceptable
Tolerance up to 850 C (1562 F)	Unacceptable
Tolerance above 850 C (1562 F)	Unacceptable

Geo-economic data for principal component

Annual world production, principal component	1.24e5	-	1.37e5	tonne/yr
Reserves, principal component	3.09e6	-	3.41e6	tonne

Primary material production: energy, climate change and water

Climate change (CO2-eq) (virgin grade production)	* 5.88	-	6.49	kg/kg
Embodied energy (virgin grade production)	* 119	-	132	MJ/kg
Water usage	* 26.6	-	29.4	l/kg

Material processing: energy

Polymer molding (injection, blow), embodied energy	* 18	-	24.8	MJ/kg

Machining, coarse, embodied energy	* 20	-	24	MJ/kg
Machining, fine, embodied energy	* 30.9	-	39.8	MJ/kg
Machining, grinding, embodied energy	* 41.7	-	55.7	MJ/kg

Material processing: climate change

Polymer molding (injection, blow), climate change (CO2-eq)	* 1.21	-	1.75	kg/kg
Machining, coarse, climate change (CO2-eq)	* 1.31	-	1.57	kg/kg
Machining, fine, climate change (CO2-eq)	* 2.02	-	2.61	kg/kg
Machining, grinding, climate change (CO2-eq)	* 2.73	-	3.65	kg/kg

Recycling and end of life

Recycle	✗			
Functional recycle	✗			
Combust for energy recovery	✓			
Heat of combustion (net)	* 30	-	31.5	MJ/kg
Combustion CO2	* 2.42	-	2.54	kg/kg
Landfill	✓			
Biodegrade	✗			
Toxicity rating	Non-toxic			
A renewable resource?	✗			

Environmental notes

Both resin and hardener are irritants; their vapors are potentially toxic. Ventilation and skin protection are important, but both are achievable. Thermosets cannot be recycled, though it may be possible to use them as fillers. Cutting and machining of glass and carbon-fiber composites requires special forced-air ventilation to remove the fine glass or carbon dust that is damaging if inhaled.

Supporting information

Design guidelines

Epoxy molding compounds are supplied in liquid or granular form. They can be shaped by transfer molding at low pressures (350-700 kPa). When designing with epoxy, as with any thermosetting material, allowance must be made for shrinkage on cooling; perfectly flat molded surfaces are not achievable, and the minimum wall thickness for average-sized epoxy molded parts is 2.0mm. Unmodified epoxies have a high viscosity; they are shaped by transfer molding. Diluted epoxy resins have a lower viscosity and cure slowly, but can be cast or used to impregnate a mat or weave of fibers. The addition of fillers gives epoxies improved machinability, hardness, impact resistance and thermal conductivity; thermal expansion and mold shrinkage are both reduced. Plasticizers and flexibilizers increase flexibility and toughness. Epoxy is also commonly used as a pattern or mold material. Epoxy resin laminates are formed using a wide range of processes, from batch techniques such as hand lay up and bulk molding compound (BMC) molding, producing, for example, mechanical components like gears and distributor caps, to continuous processes such as filament winding, pultrusion and continuous laminating, making rods or girder stock. Epoxy resins are tougher than polyesters and have lower shrinkage, but are more expensive. if brought into contact with the skin, epoxies can cause skin irritations.

Technical notes

Most epoxies are formed by the combination of bisphenol-A and epichlorohydrin in the presence of a catalyst. Catalysts include several amines and acid anhydrides, and the temperature at which the epoxy will cure (ranging from room- to high -temperature) is determined by the type of catalyst, which also affects the properties of the final product.

Typical uses

Pure epoxy molding compounds: the encapsulation of electrical coils and electronics components.
Epoxy resins in laminates: pultruded rods, girder stock, special tooling fixtures, mechanical components such as gears, adhesives, often for high-strength bonding of dissimilar materials, patterns and molds for shaping thermoplastics.

Tradenames

Araldite, Epikote, Epolite, Fiberite, Lytex, Stycast

Links

ProcessUniverse

Producers

Reference